EC-Database System Lab

BS (CE)-2023

Project Report

Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	**5**	**5**	**5**	**15**
Obtained				

Lab Instructor.
Engr. Dur-e-Shahwar

Pakistan Tourism Management System

Introduction:

The aim of this project was to design and implement a Database Management System (DBMS) for a tourism management system, named "Pakistan Tourism Management System" (PTMS). This system focuses on managing tourist data, bookings, accommodations, and guided tours, ensuring efficient organization and retrieval of information. The project incorporates database design, normalization, and the development of a user-friendly portal for CRUD operations.

Database Design:

1. Database Overview

The database includes 17 interconnected tables covering key components like tourists, bookings, packages, accommodations, and more. This structure allows robust data management, query capabilities, and scalability for future additions.

2. Tables and Relationships

Key tables include:

Tourists: Stores tourist details.

TouristPackages: Defines available packages and destinations.

Bookings: Captures bookings made by tourists.

Accommodations, Guides, Transportation: Handle related services.

Feedback and Payments: Store tourist reviews and financial transactions.

And 9 more...

Relationships:

One-to-Many: A tourist can make multiple bookings.

Many-to-Many: Destinations can have multiple guides and vice versa, managed via a junction table.

3. Normalization Process

The database was normalized to 3NF:

1NF: Eliminated multi-valued attributes (e.g., splitting ContactInfo in the Tourists table).

2NF: Removed partial dependencies (e.g., separating TotalAmount into a new table in Bookings).

3NF: Eliminated transitive dependencies (e.g., moving Event Costs to a separate table).

Refer to the provided SQL scripts for normalized table creation and data insertion.

Portal Development:

1. Technology Used

The portal was developed using Python and the Tkinter framework for a Windows-based GUI. The backend is connected to the SQL database.

2. User Interface

The portal offers:

Data Display: Listbox for browsing records.

Form-Based Inputs: Adding and updating records.

Navigation: Buttons for key actions like adding, deleting, and viewing details.

3. Features

Aesthetic design with modern fonts and color schemes.

Simulated records for testing functionality.

Integration with the database for dynamic operations.

• SQL FOR NORMALIZED DATABASE:

```sql
CREATE DATABASE PTMS1;
GO

USE PTMS1;
GO
CREATE TABLE Tourists (
    TouristID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    ContactInfo VARCHAR(100) NOT NULL,
    Nationality VARCHAR(50) NOT NULL
);

CREATE TABLE TouristPackages (
    PackageID INT NOT NULL,
    PackageName VARCHAR(100) NOT NULL,
    Destination VARCHAR(100) NOT NULL,
    Price DECIMAL(10, 2) NOT NULL,
    PRIMARY KEY (PackageID, Destination)
);

CREATE TABLE Bookings (
    BookingID INT PRIMARY KEY,
    TouristID INT NOT NULL,
    PackageID INT NOT NULL,
    Destination VARCHAR(100) NOT NULL,
    TransportID INT NULL,
    BookingDate DATE NOT NULL,
    FOREIGN KEY (TouristID) REFERENCES Tourists(TouristID),
    FOREIGN KEY (PackageID, Destination) REFERENCES TouristPackages(PackageID, Destination)
);
```

```sql

CREATE TABLE PackagePricing (
    PackageID INT NOT NULL,
    Destination VARCHAR(100) NOT NULL,
    TotalAmount DECIMAL(10, 2) NOT NULL,
    PRIMARY KEY (PackageID, Destination),
    FOREIGN KEY (PackageID, Destination) REFERENCES TouristPackages(PackageID, Destination)
);

CREATE TABLE Destinations (
    DestinationID INT PRIMARY KEY,
    DestinationName VARCHAR(100) NOT NULL,
    Region VARCHAR(50) NOT NULL,
    TouristSpots VARCHAR(50) NOT NULL
);

CREATE TABLE Guides (
    GuideID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Experience VARCHAR(50) NOT NULL,
    Language VARCHAR(50) NOT NULL
);

CREATE TABLE GuideDestinations (
    DestinationID INT NOT NULL,
    GuideID INT NOT NULL,
    PRIMARY KEY (DestinationID, GuideID),
    FOREIGN KEY (DestinationID) REFERENCES Destinations(DestinationID),
    FOREIGN KEY (GuideID) REFERENCES Guides(GuideID)
);

```

```sql

CREATE TABLE Accommodations (
    AccommodationID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Location VARCHAR(100) NOT NULL,
    CostPerNight DECIMAL(10, 2) NOT NULL
);

CREATE TABLE TouristFeedback (
    FeedbackID INT PRIMARY KEY,
    TouristID INT NOT NULL,
    BookingID INT NOT NULL,
    FeedbackText VARCHAR(MAX) NOT NULL,
    FOREIGN KEY (TouristID) REFERENCES Tourists(TouristID),
    FOREIGN KEY (BookingID) REFERENCES Bookings(BookingID)
);

CREATE TABLE Transportation (
    TransportID INT PRIMARY KEY,
    Type VARCHAR(50) NOT NULL,
    CompanyName VARCHAR(100) NOT NULL,
    Cost DECIMAL(10, 2) NOT NULL
);

CREATE TABLE TouristPayments (
    PaymentID INT PRIMARY KEY,
    TouristID INT NOT NULL,
    BookingID INT NOT NULL,
    PaymentDate DATE NOT NULL,
    Amount DECIMAL(10, 2) NOT NULL,
    FOREIGN KEY (TouristID) REFERENCES Tourists(TouristID),
```

```sql
        FOREIGN KEY (BookingID) REFERENCES Bookings(BookingID)
    );

    CREATE TABLE CulturalEvents (
        EventID INT PRIMARY KEY,
        EventName VARCHAR(100) NOT NULL,
        Location VARCHAR(100) NOT NULL
    );

    CREATE TABLE EventCosts (
        EventID INT PRIMARY KEY,
        Cost DECIMAL(10, 2) NOT NULL,
        FOREIGN KEY (EventID) REFERENCES CulturalEvents(EventID)
    );

    CREATE TABLE EmergencyContacts (
        ContactID INT PRIMARY KEY,
        DestinationID INT NOT NULL,
        Phone VARCHAR(15) NOT NULL,
        FOREIGN KEY (DestinationID) REFERENCES Destinations(DestinationID)
    );

    CREATE TABLE ContactNames (
        DestinationID INT PRIMARY KEY,
        ContactName VARCHAR(100) NOT NULL,
        FOREIGN KEY (DestinationID) REFERENCES Destinations(DestinationID)
    );

    CREATE TABLE TouristGroups (
        GroupID INT PRIMARY KEY,
        TouristID INT NOT NULL,
        GroupName VARCHAR(100) NOT NULL,
        FOREIGN KEY (TouristID) REFERENCES Tourists(TouristID)
    );

    CREATE TABLE RoomBookings (
        RoomBookingID INT PRIMARY KEY,
```

```sql
    CREATE TABLE RoomBookings (
        RoomBookingID INT PRIMARY KEY,
        AccommodationID INT NOT NULL,
        TouristID INT NOT NULL,
        StartDate DATE NOT NULL,
        EndDate DATE NOT NULL,
        FOREIGN KEY (AccommodationID) REFERENCES Accommodations(AccommodationID),
        FOREIGN KEY (TouristID) REFERENCES Tourists(TouristID)
    );
    GO

    -- ==============================
    -- Insert Sample Data
    -- ==============================
    INSERT INTO Tourists VALUES
    (1,'Ali Khan','ali.khan@example.com','Pakistani'),
    (2,'John Smith','john.smith@example.com','British'),
    (3,'Maria Lopez','maria.lopez@example.com','Spanish');

    INSERT INTO TouristPackages VALUES
    (1,'Northern Adventure','Hunza',20000),
    (2,'Cultural Tour','Lahore',15000),
    (3,'Beach Vacation','Karachi',18000);

    INSERT INTO Bookings VALUES
    (1,1,1,'Hunza',1,'2025-01-01'),
    (2,2,2,'Lahore',2,'2025-01-05'),
    (3,3,3,'Karachi',3,'2025-01-10');

    INSERT INTO PackagePricing VALUES
    (1,'Hunza',25000),
    (2,'Lahore',17000),
    (3,'Karachi',20000);

    INSERT INTO Destinations VALUES
    (1,'Hunza','Northern Pakistan','Mountains'),
    (2,'Lahore','Punjab','Historical Sites'),
    (3,'Karachi','Sindh','Beaches');
```

```sql
(2,'Lahore Inn','Lahore',3000),
(3,'Beach Resort','Karachi',4000);

INSERT INTO TouristFeedback VALUES
(1,1,1,'Amazing experience in Hunza!'),
(2,2,2,'Loved the cultural tour in Lahore.'),
(3,3,3,'Beach vacation was relaxing.');

INSERT INTO Transportation VALUES
(1,'Bus','Northern Travels',1500),
(2,'Train','Pakistan Railways',1000),
(3,'Plane','PIA',8000);

INSERT INTO TouristPayments VALUES
(1,1,1,'2025-01-02',25000),
(2,2,2,'2025-01-06',17000),
(3,3,3,'2025-01-11',20000);

INSERT INTO CulturalEvents VALUES
(1,'Music Festival','Lahore'),
(2,'Art Exhibition','Karachi'),
(3,'Food Carnival','Hunza');

INSERT INTO EventCosts VALUES
(1,5000),(2,4000),(3,3000);

INSERT INTO EmergencyContacts VALUES
(1,1,'0345-1234567'),
(2,2,'0300-9876543'),
(3,3,'0321-5678901');

INSERT INTO ContactNames VALUES
(1,'Zahid Ali'),
(2,'Ayesha Khan'),
(3,'Bilal Ahmed');

INSERT INTO TouristGroups VALUES
(1,1,'Northern Explorers'),
(2,2,'Cultural Enthusiasts'),
```

```sql
(3,3,'Beach Lovers');

INSERT INTO RoomBookings VALUES
(1,1,1,'2025-01-01','2025-01-03'),
(2,2,2,'2025-01-05','2025-01-07'),
(3,3,3,'2025-01-10','2025-01-12');
GO

-- ==============================
-- Create Full Tourism View
-- ==============================
CREATE VIEW FullTourismView AS
SELECT
    t.TouristID,
    t.Name AS TouristName,
    t.ContactInfo,
    t.Nationality,
    tp.PackageName,
    tp.Destination,
    tp.Price AS PackagePrice,
    b.BookingDate,
    p.TotalAmount AS PackageTotalAmount,
    d.DestinationName,
    d.Region,
    d.TouristSpots,
    a.Name AS AccommodationName,
    a.Location AS AccommodationLocation,
    a.CostPerNight,
    g.Name AS GuideName,
    g.Experience AS GuideExperience,
    g.Language AS GuideLanguage,
    tf.FeedbackText AS TouristFeedback,
    tr.Type AS TransportType,
    tr.CompanyName AS TransportCompany,
    tr.Cost AS TransportCost,
    pmt.PaymentDate,
    pmt.Amount AS PaymentAmount,
    ce.EventName AS CulturalEventName,
    ce.Cost AS EventCost
```

```sql
        g.Language AS GuideLanguage,
        tf.FeedbackText AS TouristFeedback,
        tr.Type AS TransportType,
        tr.CompanyName AS TransportCompany,
        tr.Cost AS TransportCost,
        pmt.PaymentDate,
        pmt.Amount AS PaymentAmount,
        ce.EventName AS CulturalEventName,
        ec.Cost AS EventCost,
        em.Phone AS EmergencyContactPhone,
        cn.ContactName AS EmergencyContactName,
        tg.GroupName AS TouristGroupName,
        rb.StartDate AS RoomBookingStart,
        rb.EndDate AS RoomBookingEnd
    FROM Tourists t
    JOIN Bookings b ON t.TouristID = b.TouristID
    JOIN TouristPackages tp ON b.PackageID = tp.PackageID AND b.Destination = tp.Destination
    JOIN PackagePricing p ON tp.PackageID = p.PackageID AND tp.Destination = p.Destination
    JOIN Destinations d ON tp.Destination = d.DestinationName
    LEFT JOIN Accommodations a ON d.DestinationName = a.Location
    LEFT JOIN GuideDestinations gd ON d.DestinationID = gd.DestinationID
    LEFT JOIN Guides g ON gd.GuideID = g.GuideID
    LEFT JOIN TouristFeedback tf ON b.BookingID = tf.BookingID
    LEFT JOIN Transportation tr ON b.TransportID = tr.TransportID
    LEFT JOIN TouristPayments pmt ON b.BookingID = pmt.BookingID
    LEFT JOIN CulturalEvents ce ON ce.Location = d.DestinationName
    LEFT JOIN EventCosts ec ON ce.EventID = ec.EventID
    LEFT JOIN EmergencyContacts em ON d.DestinationID = em.DestinationID
    LEFT JOIN ContactNames cn ON d.DestinationID = cn.DestinationID
    LEFT JOIN TouristGroups tg ON t.TouristID = tg.TouristID
    LEFT JOIN RoomBookings rb ON t.TouristID = rb.TouristID;
    GO

    -- Query the view
    SELECT * FROM FullTourismView;
```

OUTPUT:

90 % No issues found Ln: 279, Ch: 1 SPC CRLF Windows 1252

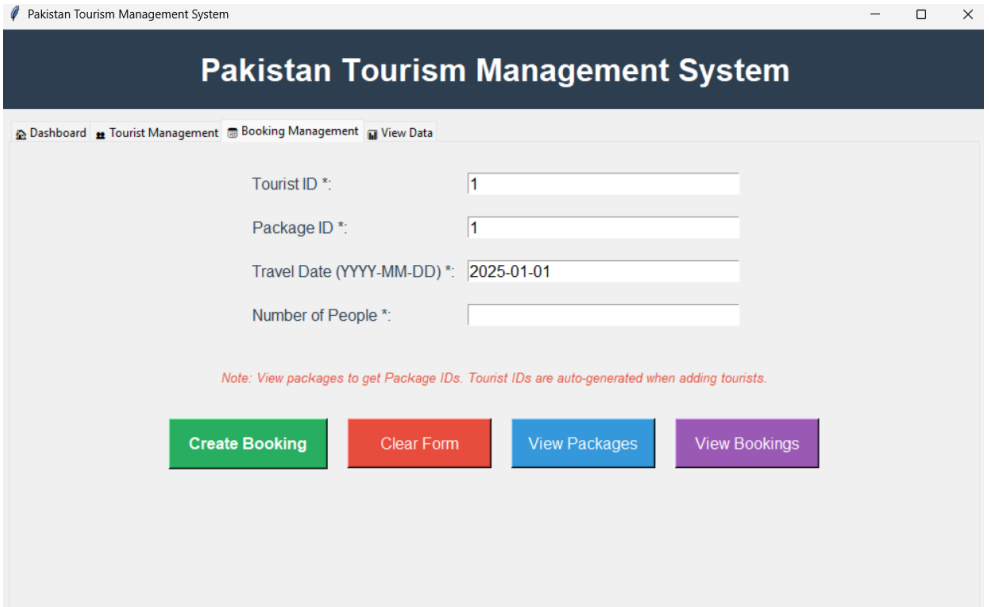
Results Messages

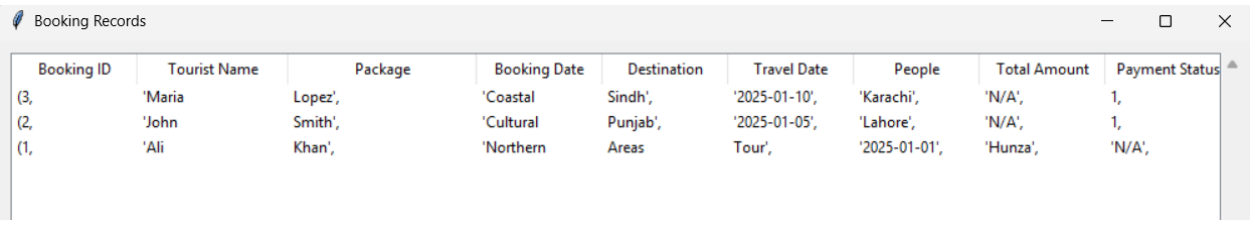
	TouristID	TouristName	ContactInfo	Nationality	PackageName	Destination	PackagePrice	BookingDate	PackageTotalAmount	DestinationName	Region	TouristSpots	AccommodationName	Accomm
1	1	Ali Khan	ali.khan@example.com	Pakistani	Northern Adventure	Hunza	20000.00	2025-01-01	25000.00	Hunza	Northern Pakistan	Mountains	Hunza Hotel	Hunza
2	2	John Smith	john.smith@example.com	British	Cultural Tour	Lahore	15000.00	2025-01-05	17000.00	Lahore	Punjab	Historical Sites	Lahore Inn	Lahore
3	3	Maria Lopez	maria.lopez@example.com	Spanish	Beach Vacation	Karachi	18000.00	2025-01-10	20000.00	Karachi	Sindh	Beaches	Beach Resort	Karachi

TASK 2 AND 3: For backend we had used python:

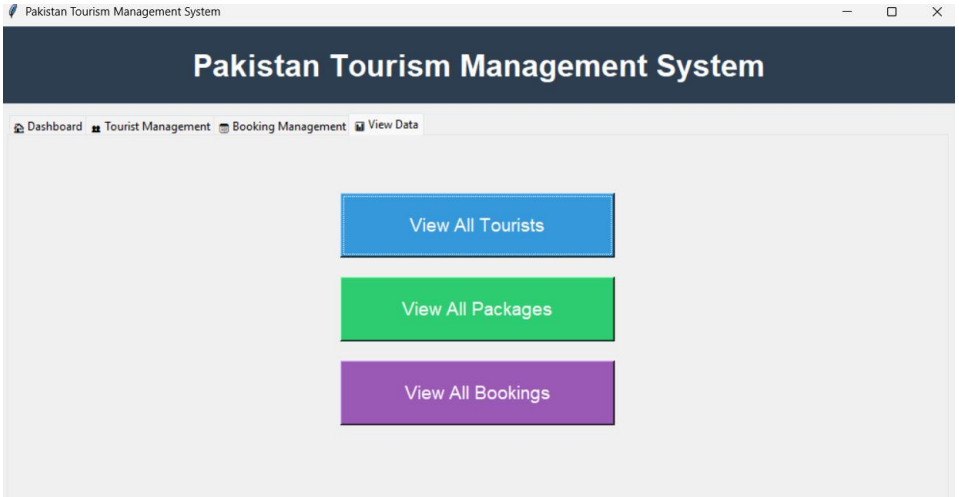
PYTHON GUI RESULT:

ID	Name	Contact	Nationality	Email	DOB	Gender	Reg Date
(1,	'Ali	Khan',	'ali.khan@exampl	'Pakistani',	'N/A',	'N/A',	'N/A',
(2,	'John	Smith',	'john.smith@exar	'British',	'N/A',	'N/A',	'N/A',
(3,	'Maria	Lopez',	'maria.lopez@exa	'Spanish',	'N/A',	'N/A',	'N/A',
(4,	'urooj',	'88999',	'pakistani',	'N/A',	'N/A',	'N/A',	'N/A')
(5,	'ali	hassan',	'9000000000',	'kurd',	'N/A',	'N/A',	'N/A',

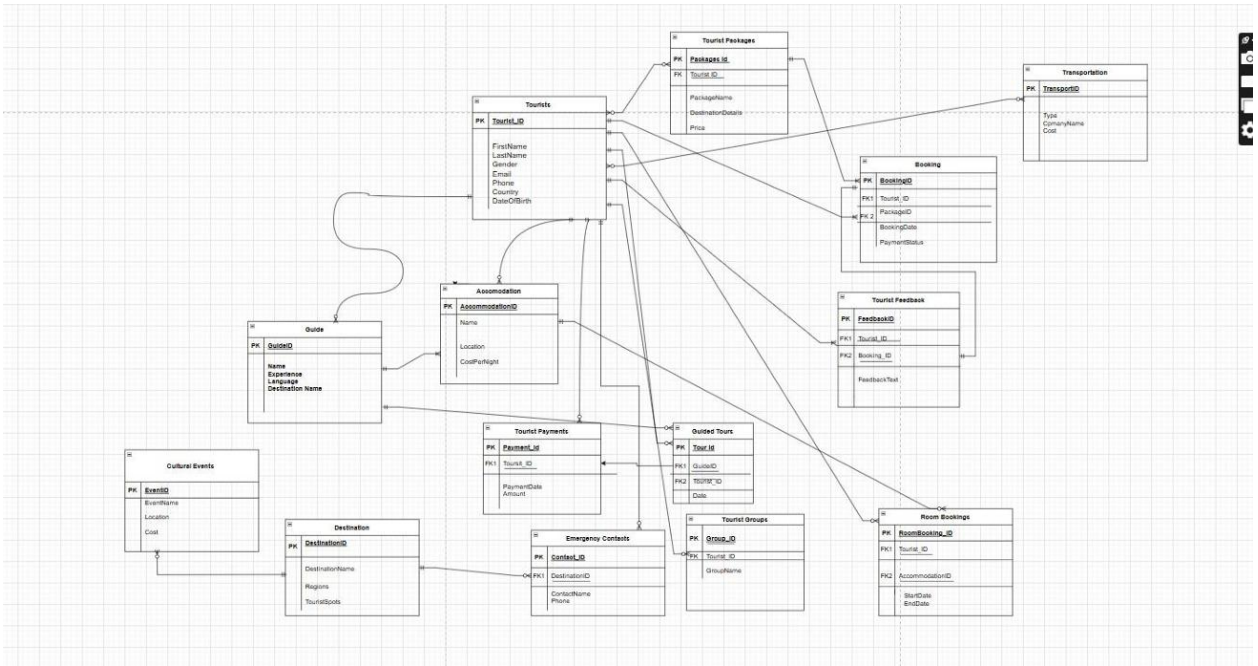




Booking ID	Tourist Name	Package	Booking Date	Destination	Travel Date	People	Total Amount	Payment Status
(3,	'Maria	Lopez',	'Coastal	Sindh',	'2025-01-10',	'Karachi',	'N/A',	1,
(2,	'John	Smith',	'Cultural	Punjab',	'2025-01-05',	'Lahore',	'N/A',	1,
(1,	'Ali	Khan',	'Northern	Areas	Tour',	'2025-01-01',	'Hunza',	'N/A',



ERD:



CONCLUSION:

The Pakistan Tourism Management System is a robust, user-friendly DBMS that meets the specified requirements. It showcases the integration of database design principles, normalization, and functional CRUD operations in a real-world application. The portal enhances usability and ensures efficient data management.